

Build 4: Agent Testing Log

Jacob Poore and Arian Zarazua

April 24, 2026

Overview

We tested the Build 4 HITL Router Agent using a game-stats CSV (5,980 rows, NFL seasons 2013–2023, 35 columns) covering per-game offensive statistics for all 32 NFL teams. We ran 7 prompts total: 4 passed and 3 failed.

Prompt Log

#	Route	Result	Notes
1	Tool	Pass	Correct tool and args
2	Tool	Fail	No team filter; aggregated all teams
3	Codegen	Fail	Wrong team names; chart is empty
4	Tool	Pass	Summarized market column correctly
5	Codegen	Fail	Summed a string column; axis labels malformed
6	Tool	Pass	Heatmap saved correctly
7	Tool	Fail	<code>plot_missingness()</code> argument conflict error

Table 1: Summary of prompts. Prompts: (1) plot avg passing yards by season; (2–3) plot NFC East rushing and passing yards by season; (4) summarize a random categorical variable; (5) plot market value by team for 2023; (6) correlation heatmap for offensive metrics; (7) missingness patterns across performance metrics.

Test Details

Test 1 – Passing yards over seasons [Pass]

Prompt: Plot average passing yards by season.

The router selected `plot_temporal_line_chart` with the correct arguments. The chart shows mean passing yards per game ranging from about 235 to 259 yards across 2013–2023, with a dip in 2017 and a decline after 2021.

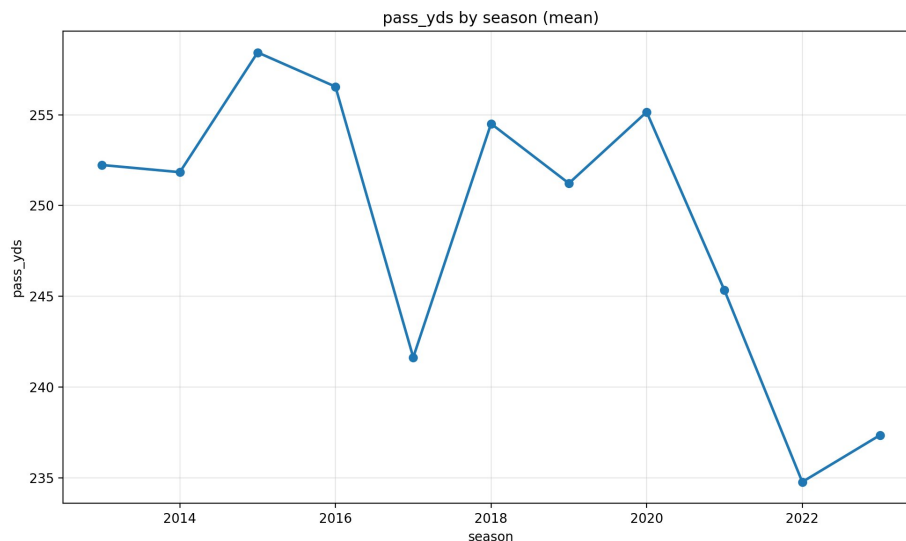


Figure 1: Mean passing yards per game by season (2013–2023).

Test 2 – NFC East yards by season via tool route [Fail]

Prompt: Plot total rushing and passing yards for every team in the NFC East by season.

The router sent this to `aggregate_by_temporal_column` with `aggregation_method = sum`. The tool returned 11 season groups but aggregated the entire dataset with no team filter. The output shows league-wide totals rather than per-team NFC East figures.

What went wrong: The tool does not support filtering by team or division. This request needed codegen so a filter on the `alias` column could be applied first.

Test 3 – NFC East yards by season via codegen [Fail]

Prompt: (follow-up via `code` command) Plot total rushing and passing yards for every team in the NFC East by season.

The agent generated a script filtering on `alias` using full team names (e.g., "Dallas Cowboys"). The script exited with code 0, but the chart is blank. The `alias` column uses abbreviated identifiers, so no rows matched the filter.

What went wrong: The agent did not check what values `alias` actually contains before writing the filter. Running `summarize_categorical` on `alias` first would have revealed the correct identifiers.

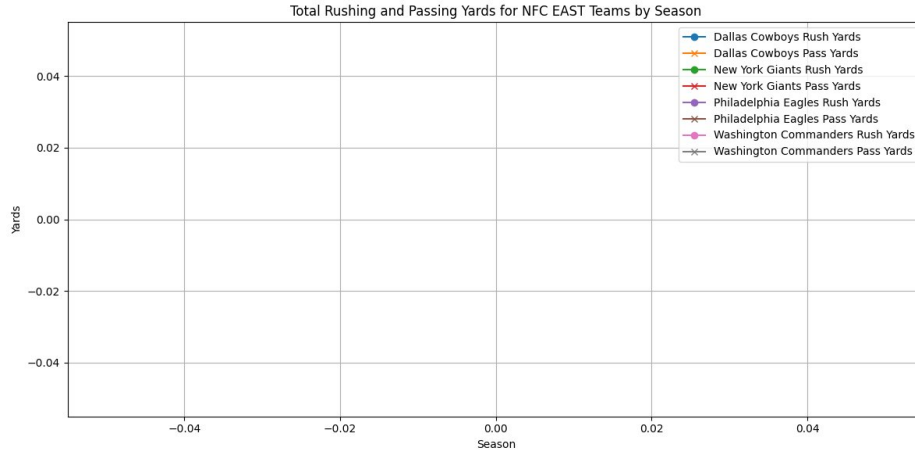


Figure 2: NFC East rushing and passing yards by season. The chart is empty because the team name filter matched zero rows.

Test 4 – Summarize categorical variable [Pass]

Prompt: Summarize a random categorical variable.

The router selected `summarize_categorical` and chose the `market` column. Results: 5,980 entries, 0 missing, 30 unique markets. Top values were Los Angeles (374) and New York (361).

Test 5 – Market value by team for 2023 [Fail]

Prompt: Plot market value by team for 2023 data.

The agent generated code that filtered to `season == 2023`, grouped by `alias`, and called `.sum()` on the `market` column. The script exited cleanly, but `market` is a string column containing city names. Summing it concatenated the strings, producing axis labels like `ArizonaArizona` and `DallasDallasDallas`.

What went wrong: No numeric market-value column exists in the dataset. The agent should have flagged the missing column and asked for clarification instead of running code on the wrong column type.



Figure 3: Attempted market value by team for 2023. The `market` column holds city name strings; summing it produced concatenated labels and an unreadable chart.

Test 6 – Correlation heatmap [Pass]

Prompt: Show a correlation heatmap for offensive metrics.

The router selected `plot_corr_heatmap` and the tool ran correctly across all 27 numeric columns. Strong positive correlations appear among passing volume metrics (`pass_att`, `pass_cmp`, `pass_yds`) and between `penalties` and `penalty_yds`. `pass_int` shows a moderate negative correlation with `passer_rating`.

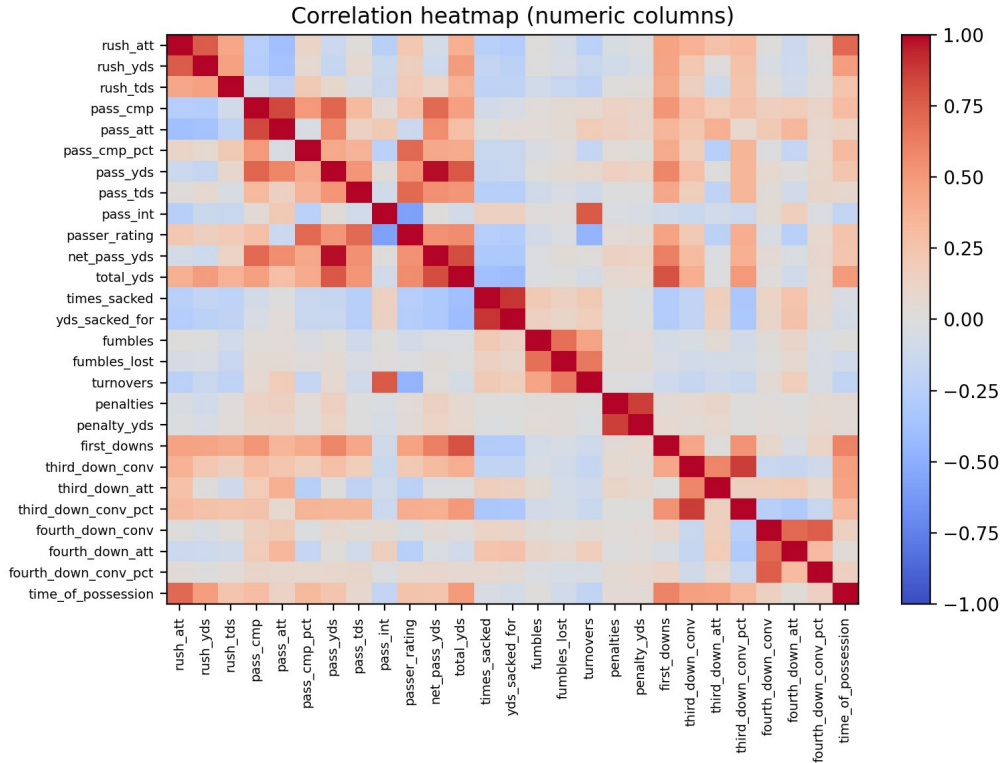


Figure 4: Correlation heatmap across all 27 numeric columns.

Test 7 – Missingness patterns [Fail]

Prompt: What patterns of missingness exist in the performance metrics across different seasons?

The router selected `plot_missingness` and the user approved. The tool threw:

ERROR running tool: `plot_missingness()` got multiple values for argument 'miss_df'

What went wrong: The router passed `miss_df` as both a positional and keyword argument, causing a `TypeError`. The router also passed an empty dict rather than a computed missingness DataFrame. This is the same argument-construction bug seen in Build 3 (Tests 4 and 9).

Langfuse Tracing Verification

Langfuse tracing was active throughout the session. The export contains 31 trace spans covering every agent step: `build-router`, `build-run-tool`, `build-summarize`, `build-codegen`, and `build-suggest`. The Langfuse router traces confirm the RAG output was used.

RAG chunk retrieval – representative traces

The following are four representative RAG retrievals from the session, drawn directly from terminal output.

Trace: plot average passing yards by season

Langfuse span id: 95289029c24185c40b809e4180758d81 (18:36:46 UTC)

```
[Chunk 1 | source=nfl_analysis_guides\basic_nfl_data_analysis_workflow.md  
          | section=Step 2: map to dataset | score=0.5769]  
Stats, Snapcounts, Scoring, ExpectedPoints, Seasons, Roster.
```

```
[Chunk 2 | source=datasets_notes\Seasons\season_analysis_workflow.md  
          | section=Step 1: aggregate games | score=0.5057]  
team-season grouping.
```

```
[Chunk 3 | source=nfl_analysis_guides\basic_nfl_data_analysis_workflow.md  
          | section=Step 5: compute summaries | score=0.5151]  
production, efficiency, turnovers, usage.
```

Router output consumed from RAG context:

```
{"mode":"tool","tool":"plot_temporal_line_chart",  
  "args":{"temporal_column":"season","numeric_column":"pass_yds"}}
```

Trace: NFC East rushing and passing yards by season

Langfuse span id: d043a6ad50e18cbe9a8e1caa0745a4d2 (18:37:56 UTC)

```
[Chunk 1 | source=nfl_analysis_guides\basic_nfl_data_analysis_workflow.md  
          | section=Step 2: map to dataset | score=0.5333]  
Stats, Snapcounts, Scoring, ExpectedPoints, Seasons, Roster.
```

```
[Chunk 2 | source=datasets_notes\Seasons\season_analysis_workflow.md  
          | section=Step 1: aggregate games | score=0.5173]  
team-season grouping.
```

```
[Chunk 3 | source=nfl_analysis_guides\team_strength_analysis.md  
          | section=Step 1: aggregate season or games | score=0.4854]  
group by team.
```

Router output consumed from RAG context:

```
{"mode":"tool","tool":"aggregate_by_temporal_column",  
  "args":{"temporal_column":"season",  
          "numeric_columns":["rush_yds","pass_yds"],  
          "aggregation_method":"sum"}}
```

Trace: summarize a random categorical variable

Langfuse span id: 7b6409f89e6b43ffedde3e1ad0caff34 (18:42:26 UTC)

```
[Chunk 1 | source=build0_tool_notes\summarize_categorical.md  
          | section=Purpose | score=0.5464]
```

Summarize categorical variables using counts and common values.

```
[Chunk 2 | source=build0_tool_notes\summarize_categorical.md  
          | section=Inputs | score=0.5365]
```

- df - cat_cols - optional top_k for most frequent categories

```
[Chunk 3 | source=analysis_guides\tool_selection_rules.md  
          | section=Dataset overview requests | score=0.5331]
```

If the user asks to describe or summarize the dataset,
start with a basic profile and variable-type split.

Router output consumed from RAG context:

```
{"mode":"tool","tool":"summarize_categorical",  
  "args":{"column":"market"}}
```

Trace: missingness patterns across performance metrics

Langfuse span id: cbedcc2e1e34a07ccb43fe34831d4aa4 (18:41:12 UTC)

```
[Chunk 1 | source=nfl_analysis_guides\basic_nfl_data_analysis_workflow.md  
          | section=Step 2: map to dataset | score=0.5302]
```

Stats, Snapcounts, Scoring, ExpectedPoints, Seasons, Roster.

```
[Chunk 2 | source=datasets_notes\Roster\depth_chart_reasoning.md  
          | section=Step 4: usage trends | score=0.5112]
```

changes over season.

```
[Chunk 3 | source=datasets_notes\Seasons\season_analysis_workflow.md  
          | section=Step 4: consistency | score=0.5099]
```

variance across games.

Router output consumed from RAG context:

```
{"mode":"tool","tool":"plot_missingness",  
  "args":{"miss_df":{},"out_path":""}}
```

Observations

RAG retrieval fired on every router call and returned 3–4 chunks per query. Chunk scores ranged from roughly 0.48 to 0.58 across all queries. The top-scoring source across most queries was `basic_nfl_data_analysis_workflow.md`, which appears to be a broadly relevant document.